The Economics of Bicycles for the Mind

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NBER Working Paper 34034

Repository 2 — Artificial Intelligence and Economic Modeling

The Paper: Cognitive Tools and Human Capabilities

Question. Why does the empirical record on cognitive technology and inequality look contradictory — computers raised productivity *and* inequality, while AI often raises productivity while *reducing* it within firms?

The single mechanism. A cognitive tool is a substitute for implementation skill s, but a complement to judgment — and judgment splits into two skills that behave differently:

Opportunity judgment γ

- Spotting a chance to improve the task
- *Always* complemented by a better tool

Payoff judgment α

- Acting correctly on a successful implementation
- Complemented only *conditionally*

Agent's problem

$$\max_{e_t \geq 0} \; M(e_t;\theta) = p(se_t;\theta)\,\alpha\,\Delta - c(e_t;\theta)$$

$$\text{FOC:} \quad p'(se_t^*;\theta)\,s\alpha\Delta = c'(e_t^*;\theta)$$

The Main Result: Proposition 1, with All Its Conditions

Definition 1 (cognitive tool)

For all e and all $\theta' > \theta$:

(i) $p(se;\theta') \geq p(se;\theta), \quad c(e;\theta') \leq c(e;\theta)$

(ii) $\dfrac{p'(se;\theta)}{c'(e;\theta)}$ strictly decreasing in θ

Proposition 1

Under Definition 1 and the standing regularity assumptions (p increasing, weakly concave; c increasing, weakly convex), as θ rises from 0 to 1:

1. $e_t^*(1) < e_t^*(0)$ for all t
2. $e_t^*(\theta) = e^*(\theta)$ for all t
3. $V_0(1) > V_0(0)$

A better tool lowers the marginal payoff of effort relative to its cost, so the agent works less per opportunity — but net benefit still rises, so does the value of the whole process.

What I Did

1. **Audited the variance-U-shape claim (Prop. 3(b), Appendix A.3).** The naive claim — "variance is U-shaped in θ, no conditions" — fails: it holds only under condition (30). But the paper's own proof text goes further: it states "therefore $B < 0$" as if condition (30) is what establishes $B < 0$. In fact $B < 0$ follows from Jensen's inequality alone, *independent* of condition (30); the real conditional fact is whether the Γ-heterogeneity term can overturn it in the full expression.

$$B = \frac{\Delta^2}{2}(\mu_\alpha^2 + \sigma_\alpha^2)\big(1 - \mu_s E[1/s]\big) \leq 0 \text{ unconditionally, yet the sign of } \left.\frac{\partial\,\mathrm{Var}(V(\theta))}{\partial\theta}\right|_{\theta=0} \text{ is not.}$$

2. **Tested Definition 1's channel symmetry (`extensions/definition-asymmetry.md`).** Isolating the precision channel (p only) gives a clean value gain θ/s. Isolating the cost channel first looked *impossible* under the paper's own normalization $c(0;\theta) = 0$ — I initially described this as failing "eventually," which was imprecise: the contradiction is immediate, at every $e > 0$. Relaxing that normalization (letting a fixed overhead shrink with θ) produced an explicit, verified construction where the cost-only channel *is* valid on the full domain.

Takeaway

Both findings share a shape: a claim that looks unconditional on first read turns out to depend on an assumption the paper (or I) didn't surface — condition (30) vs. the unconditional sign of B; the

Where I Did Not Believe the AI

What the model claimed. In auditing the variance-U-shape result (Prop. 3(b), Appendix A.3), the model asserted that, for implementation skill s with mean μ_s and any strictly positive variance $\sigma_s^2 > 0$,

$$\mu_s\, E\left[\frac{1}{s}\right] \geq 1 \qquad \Longrightarrow \qquad B = \frac{\Delta^2}{2}\left(\mu_\alpha^2 + \sigma_\alpha^2\right)\Big(1 - \mu_s E[1/s]\Big) \leq 0$$

holds unconditionally — independent of condition (30) and of any moment of Γ — with Jensen's inequality applied to the convex function $1/x$ on $(0, \infty)$ as the justification.

What I checked by hand. I reproduced the three-step chain on paper: Jensen's inequality for a strictly convex function f, verified that $1/x$ is strictly convex on $(0, \infty)$ since its second derivative $2/x^3 > 0$ there; applied it to s to get $E[1/s] \geq 1/\mu_s$ and hence $\mu_s E[1/s] \geq 1$, with equality only if s is degenerate; and substituted this into the definition of B from eq. (69), confirming the sign falls out without invoking condition (30) or any property of Γ at any step.

My verdict: correct. The claim that $B \leq 0$ unconditionally holds up exactly as stated — Jensen's inequality alone, applied to implementation-skill heterogeneity, is sufficient, and condition (30) plays no role in establishing this particular fact. What the hand derivation does *not* rescue is the paper's own proof, which uses this unconditional fact as if it were what makes the full derivative in eq. (73) negative — that inferential step is separate from, and not justified by, the claim verified here.